Example 2.44

Evaluate $\int \frac{x^2}{x^2-25}dx$

Solution:

$$\int \frac{x^2}{x^2-25}dx = \int \frac{(x^2-25)+25}{x^2-25}dx$$

$$= \int \left\{1+\frac{25}{x^2-25}\right\}dx$$

$$= \int dx + 25\int \frac{dx}{x^2-25}$$

$$= x + 25\left[\frac{1}{2(5)}\log\left|\frac{x-5}{x+5}\right|\right] + c$$

$$= x + \frac{5}{2}\log\left|\frac{x-5}{x+5}\right| + c$$

Example 2.45

Evaluate $\int \frac{dx}{x^2-3x+2}$

Solution:

By completing the squares

$$x^2-3x+2 = \left(x-\frac{3}{2}\right)^2 - \frac{9}{4} + 2$$

$$= \left(x-\frac{3}{2}\right)^2 - \frac{1}{4}$$

$$= \left(x-\frac{3}{2}\right)^2 - \left(\frac{1}{2}\right)^2$$

$$\int \frac{dx}{x^2-3x+2} = \int \frac{dx}{\left(x-\frac{3}{2}\right)^2 - \left(\frac{1}{2}\right)^2}$$

$$= \frac{1}{2\left(\frac{1}{2}\right)}\log\left|\frac{\left(x-\frac{3}{2}\right)-\frac{1}{2}}{\left(x-\frac{3}{2}\right)+\frac{1}{2}}\right| + c$$

$$= \log\left|\frac{2x-4}{2x-2}\right| + c$$

$$= \log\left|\frac{x-2}{x-1}\right| + c$$

Example 2.46

Evaluate $\int \frac{dx}{\sqrt{4x^2-9}}$

Solution:

$$\int \frac{dx}{\sqrt{4x^2-9}} = \int \frac{dx}{\sqrt{4\left[x^2-\frac{9}{4}\right]}}$$

$$= \frac{1}{2}\int \frac{dx}{\sqrt{x^2-\left(\frac{3}{2}\right)^2}}$$

$$= \frac{1}{2}\log\left|x+\sqrt{x^2-\left(\frac{3}{2}\right)^2}\right| + c$$

$$= \frac{1}{2}\log\left|x+\sqrt{x^2-\frac{9}{4}}\right| + c$$

$$= \frac{1}{2}\log\left|2x+\sqrt{4x^2-9}\right| + c$$

DO YOU KNOW?

$$\int \frac{m}{\sqrt{(mx)^2-a^2}}dx = \int \frac{d(mx)}{\sqrt{(mx)^2-a^2}}$$

$$= \log\left|mx+\sqrt{(mx)^2-a^2}\right| + c$$

Note

$m\log n \pm k = c$

Example 2.47

Evaluate $\int \frac{dx}{\sqrt{x^2-3x+2}}$

Solution:

By completing the squares

$$x^2-3x+2 = \left(x-\frac{3}{2}\right)^2 - \frac{9}{4} + 2$$

$$= \left(x-\frac{3}{2}\right)^2 - \left(\frac{1}{2}\right)^2$$

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